



# Merton Truck Company

## Teaching Note

### Introduction

The case "Merton Truck Company" (#9-189-163) introduces several key concepts in linear programming. The case's setting, while artificial, is typical of most introductory linear programming problems. Merton Truck Company manufactures two models of trucks in four manufacturing departments. The production technology is linear, so also the economics. The student must (1) determine the optimal product mix; (2) perform elementary sensitivity analyses; and (3) investigate the impact of changes in the product line (Should the company introduce a third truck model?), resource constraints (Should there be overtime operations in one of the four departments?), and sales mix requirements (What will be the impact of a decision to produce at least three times as many Model 101 trucks as Model 102 trucks?). The case is adapted from "Sherman Motor Company" (#9-107-010), which, in turn, was based on a paper by Robert Dorfman ("Mathematical or 'Linear' Programming: A Nonmathematical Approach," *American Economic Review*, December 1953).

### Assignment

Prepare: Merton Truck Company (#9-189-163)

Assignments: Prepare the problems on page 4 of the case.

### Suggested Sequence and Timing of Class Discussion

1. Problem Formulation—Conceptual (10 minutes)
  - What is it that Merton is deciding?
  - On what should the decision be based?
  - What factors constrain Merton's choices?
2. Problem Formulation—Analytical (15 minutes)
  - Decision variables
  - Objective function

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*This teaching note was prepared as an aid to instructors in the use of the case "Merton Truck Company" #9-189-163, in the classroom.*

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- Constraints

### 3. Solution

(55 minutes)

- One-model case
- Two-model case
- Sourcing out engine assembly?
- Using linear programming software
- Model 103?
- Overtime production?
- Marketing constraint?

### Suggested Blackboard Use

See Exhibit 1.

### Teaching Strategy

The problem setting in Merton Truck Company is typical of most introductory linear programming problems and the case should be relatively easy to teach. The case is "armchaired," and while there is some play around the concept of relevant costs and contribution, the analysis is by no means difficult. Even the numbers have been chosen to reduce the numerical complexity. This should facilitate a focus on the issues and concepts central to linear programming.

The case has four objectives:

1. It provides a vehicle for the students to learn about problem formulation: What are the appropriate decision variables? The objective function? The constraints?
2. It requires the students to draw a distinction between full costs (costs inclusive of overhead allocations) and relevant costs (costs that depend on the decision)—Merton's executives seem unable to make the distinction. While the case is fictitious, erroneous consideration of costs is rather commonplace in the "real world."
3. It offers a context to guide students through a complete solution of a simple linear programming problem.
4. The case discussion and the assignment questions suggest typical sensitivity analyses that can be performed with the help of a personal computer and a linear programming software package.

The instructor can begin class with a conceptual discussion of Merton's decisions. Questions that may be posed to the students are: What is it that Merton must decide? On what criterion should the decision be based? And what factors constrain Merton's decision alternatives? These questions may be answered "in words." Indeed, the instructor should insist on a coherent answer in English: it is surprising how many students get stuck at this very basic level—not because the questions are tricky, but because they do not have the necessary discipline to think systematically about problem formulation. Budget approximately ten minutes for this phase of class.

Having understood the problem at a conceptual level, the class can move on to formulating it quantitatively. Once notations are introduced for the number of trucks produced of each model, it should be easy to write down the appropriate constraints. The technological coefficients going into each constraint (machine-hours of each resource required to produce one unit of each truck model)

and the "right-hand side" values (machine-hours available of each resource) are given in **Exhibit 1** in the case. Do not forget to mention the nonnegativity constraints.

Writing the correct objective function may be more difficult. First, the students must decide whether they wish to maximize profit or contribution (it does not matter which; the optimal production mix in either case will be the same). Whatever the choice, they must obtain the correct value of contribution. Clearly, the contribution is not -\$1,205 for Model 101 and \$2,070 for Model 102—these are unit "profits" if Merton manufactures 1,000 Model 101 trucks and 1,500 Model 102 trucks; for other levels of production, the allocation of fixed costs would be different. The correct unit contribution is \$3,000 for Model 101 and \$5,000 for Model 102.

Moving on to a solution of the problem, the instructor can ask the following hypothetical question: If Merton decides to manufacture only one truck model, which model should it produce? The answer is obvious if there are no constraints: the model that yields the highest unit contribution. In the presence of production constraints, however, Merton might be better off producing a model with a lower unit contribution.

Next, if the company wishes to manufacture both models, how many units of each should it produce? It is useful to link this question with the analysis of the one-model case. There, not only could the student determine which model to manufacture and at what monthly rate, he or she could also determine the shadow prices for the four resource constraints (How much contribution would be generated [lost] by an incremental increase [decrease] of one unit of each resource?), and the degree of capacity utilization. With this information Merton can decide whether it should produce the second truck model as well, and in what quantity.

An alternative approach is to solve the problem graphically. This method is relatively straightforward, and students familiar with linear programming might already have done so.

Extending the discussion to a many-product case, it should be apparent that the graphical approach would not be very useful if there are more than two decision variables (the addition of one decision variable results in an extra dimension in the graphical analysis). Algorithmic solution techniques are available to solve such larger problems. But, in this day of personal computers and decision support software, it is not essential to expose the MBA students to simplex and other algorithms; they can simply use a linear programming software package.

It is to the use of such software that the instructor should turn to next, illustrating in "real-time" how to set up the problem, obtain the solution, and interpret the output. The students' learning about the use of the software will be reinforced by the problems at the back of the case. These require an examination of the sensitivity of the solution to variations in the problem description: What are the consequences of an additional unit of a resource? What is the maximum rent Merton should be willing to pay for additional engine assembly capacity? Should the company introduce a third truck model? Should it undertake overtime production? What will be the impact of a new marketing constraint? These questions can be answered by exploring the output of the original solution or by adapting the problem formulation and resolving the problem.

## Analysis

### 1. Problem Formulation—Conceptual

The Merton Truck Company must decide the optimal production rate (number of trucks per month) for each of its two truck models—Model 101 and Model 102. The company should maximize profit (revenue - variable costs - fixed costs) or contribution (revenue - variable costs). (Since fixed costs do not vary with the level of production, the optimal production quantities are the same in either case.) In the absence of constraints—technological, market-based, resource, institutional, or

managerial—Merton should produce infinite quantities of each model with a positive contribution. Constraints restrict the company's alternatives, leaving it with a "feasible set" of production possibilities. The challenge is to find the most attractive feasible production mix.

The optimal solution is a function of the unit contributions, production parameters ("technological coefficients"), and resource availability. Changes in each of these would impact the solution—effects that can be quantified through appropriate sensitivity analyses.

## 2. Problem Formulation—Analytical

### Decision variables

Let N101 and N102 be the monthly production rates of Model 101 trucks and Model 102 trucks.

### Objective function

Unit contribution for Model 101 trucks

= selling price - variable costs

= selling price - [direct material + direct labor + variable overhead]

= \$39,000 - \$[24,000 + 4,000 + 8,000] = \$3,000.

Similarly, unit contribution for Model 102 trucks

= \$38,000 - \$[20,000 + 4,500 + 8,500] = \$5,000.

Merton's objective function is:

Max  $(3,000 \cdot N101) + (5,000 \cdot N102)$ .

N101, N102

### Constraints

Consider the engine assembly department. The production of one unit of Model 101 and Model 102 take up, respectively, 1.0 and 2.0 hours of machine capacity. Thus, N101 Model 101 trucks and N102 Model 102 trucks would take up  $[(1.0 \cdot N101) + (2.0 \cdot N102)]$  hours of engine assembly capacity. Only 4,000 machine-hours of capacity are available each month. Thus,

$$(1.0 \cdot N101) + (2.0 \cdot N102) \leq 4,000$$

Similarly, for metal stamping we require

$$(2.0 \cdot N101) + (2.0 \cdot N102) \leq 6,000;$$

Model 101 assembly,

$$(2.0 \cdot N101) \leq 5,000;$$

and Model 102 assembly,

$$(3.0 \cdot N102) \leq 4,500.$$

Furthermore, Merton must produce nonnegative quantities of each model:

$$N101 \geq 0, \quad N102 \geq 0.$$

### The Complete Problem

|                                 | N101 | N102 |             |
|---------------------------------|------|------|-------------|
| Engine assembly                 | 1    | 2    | $\leq 4000$ |
| Metal stamping                  | 2    | 2    | $\leq 6000$ |
| Model 101 assembly              | 2    | 0    | $\leq 2500$ |
| Model 102 assembly              | 0    | 3    | $\leq 4500$ |
| Objective                       | 3000 | 5000 |             |
| PLUS nonnegativity constraints. |      |      |             |

### 3. Solution

#### One-Model Case

Suppose Merton wants to produce Model 101 or Model 102, but not both. Which model should it manufacture? Model 102 has a higher unit contribution (\$5,000 vs. \$3,000); if there were no constraints, Model 102 is what Merton should make.

But there are constraints. These restrict Model 101 production to  $\text{Min } [4000/1, 6000/2, 5000/2] = 2500$  and Model 102 production to  $\text{Min } [4000/2, 6000/2, 4500/3] = 1500$ . If the unit contribution for Model 101 exceeds  $[3/5 (=1500/2500)] \times$  the unit contribution for Model 102, then Merton would be better off producing Model 101—even though Model 101's unit contribution might be lower than that of Model 102!

In our case, the unit contributions for Model 101 and Model 102 are \$3000 and \$5000, respectively. If Merton were to produce only Model 101 trucks, the maximum contribution the company could realize is \$7.5 million. If, instead, Merton produced only Model 102 trucks, the maximum contribution would also be \$7.5 million. Since the maximum contribution is the same in either case, the company should be indifferent to producing Model 101 trucks or Model 102 trucks.

Suppose Merton decides to manufacture Model 101 trucks. It can produce up to 2500 trucks for a total contribution of \$7.5 million. At this level of production, Model 101 assembly capacity is fully used up, but there is excess engine assembly, metal stamping, and Model 102 assembly (1500, 1000, and 4500 machine-hours, respectively). If an additional hour of Model 101 assembly capacity were available, it would be worth  $[\frac{1}{2} \text{ trucks} \times \$3000/\text{truck}] = \$1500$  in additional contribution (one Model 101 truck requires 2 hours of Model 101 assembly time; therefore, Merton can manufacture  $\frac{1}{2}$  a truck in 1 hour of additional Model 101 assembly time). Similarly, a decrease of one hour in Model 101 assembly capacity will cost the company \$1500 in lost contribution. Thus, the "shadow price" for the Model 101 assembly constraint is \$1500. The shadow prices for engine assembly, metal stamping, and Model 102 assembly are zero—there is excess capacity in each department, and an incremental change in capacity will have no impact on Merton's contribution.

#### Two-Model Case

Should Merton also manufacture Model 102 trucks? Clearly, it should: Model 102 has a positive unit contribution and it only takes up capacity in the departments that are currently not fully used up. The additional contribution generated by producing one Model 102 truck (over and above 2500 Model 101 trucks) equals  $[(1 \times 5000) - (2 \times 0) - (2 \times 0) - (3 \times 0)] = \$5000$  (unit contribution less sum of resource requirements times shadow price for the resource). How many Model 102 trucks should Merton produce?

In the one-model case, we saw that the excess capacities in engine assembly, metal stamping, and Model 102 assembly were 1500, 1000, and 4500 machine-hours, respectively. The maximum number of Model 102 trucks that can be manufactured with these excess capacities equals  $\text{Min}[1500/2, 1000/2, 4500/3] = 500$ . Thus, the additional contribution for Merton would be  $500 \times \$5000 = \$2.5$  million.

At this stage, Merton produces 2500 Model 101 trucks and 500 Model 102 trucks per month for a total contribution of \$10.0 million. Metal stamping and Model 101 capacities are fully used up, while there is excess capacity in engine assembly and Model 102 assembly.

Can the company improve upon this? Yes. To see this, observe that producing one more Model 102 truck (over and above 500) requires Merton to produce one less Model 101 truck—otherwise, the metal stamping constraint would be violated. Is this trade-off worthwhile? Yes, because the company's contribution will increase by  $5000 - 3000 = \$2000$ . How many additional Model 102 trucks should Merton make?

500. When the company was producing 2500 Model 101 trucks and 500 Model 102 trucks per month, the excess capacities in engine assembly and Model 102 assembly were 500 and 3000 machine-hours, respectively. Producing one more Model 102 truck and, at the same time, one less Model 101 truck (which the running-at-full-capacity metal stamping requires) would take up an incremental hour  $(= 2 - 1)$  of engine assembly capacity and three hours of Model 102 capacity. Hence, the additional number of Model 102 trucks that can be manufactured equals  $\text{Min}[500/1, 3000/3] = 500$ .

Now, Merton produces  $2500 - 500 = 2000$  Model 101 trucks and  $500 + 500 = 1000$  Model 102 trucks. The company's contribution is \$11 million. Engine assembly and metal stamping are running at capacity while there is excess capacity in Model 101 and Model 102 assembly departments.

Can Merton do even better? No. Any further increase in Model 102 production can only be achieved through a reduction of two Model 101 trucks for every additional Model 102 truck (the running-at-full-capacity engine assembly department requires this). The net impact on contribution is  $[-(2 \times 3000) + (1 \times 5000)] = -\$1000$ . In other words, the contribution will go down. Therefore, the company should not further increase Model 102 truck production at the expense of Model 101 trucks.

The "optimal" solution, then, is: 2000 Model 101 trucks and 1000 Model 102 trucks per month—for a total contribution of \$11 million. At these production levels, the engine assembly and metal stamping departments are operating at capacity, while there is "slack" (excess capacity) in Model 101 assembly (1000 machine-hours  $[= 5000 - 2 \times 2000]$ ) and Model 102 assembly (1500 machine-hours  $[= 4500 - 3 \times 1000]$ ). The shadow prices for the two assembly departments are zero (since these have excess capacities, Merton will not benefit by having additional units of capacity in these departments). To find the shadow price of the engine assembly constraint, consider the consequences of an additional machine-hour of engine assembly capacity. The metal stamping constraint requires the company to reduce Model 101 production by one truck for every additional Model 102 truck that is produced. At the same time, the manufacture of one less Model 101 truck and one more Model 102 truck requires a net incremental one  $(= 2 - 1)$  hour of engine assembly time—exactly the amount of additional engine assembly capacity available. We conclude that Merton can use the additional hour of engine assembly capacity to produce one more Model 102 truck at the expense of one Model 101 truck, for a net impact on contribution of \$2000  $(= \$5000 - \$3000)$ . The shadow price of the engine assembly constraint is, therefore, \$2000. How many units of additional engine assembly capacity can Merton use up—before running out of capacity in some other department? 500 machine-hours  $(= 1500 \text{ machine-hours' slack in Model 102 assembly} \div 3 \text{ machine-hours/Model 102 truck assembled})$ .

We can adopt the same approach to determine the shadow price of the metal stamping constraint. Suppose the company had an extra machine-hour of metal stamping capacity. The engine assembly constraint requires Merton to reduce Model 102 production by one truck for every two

additional Model 101 trucks produced. The net impact on metal stamping requirement for such a change in production mix is 2 hours ( $2 \times 2 - 1 \times 2$ ). It follows that if Merton had one extra hour of metal stamping capacity, it would produce one more Model 101 truck and  $\frac{1}{2}$  less Model 102 truck, for a net impact on contribution of \$500. How many additional units of metal stamping capacity can Merton use up before running out of capacity in some other department? 500 machine-hours (= 1000 machine-hours' slack in Model 101 assembly  $\div$  2 machine-hours/Model 101 truck assembled).

Obviously, it becomes difficult to keep track of all the trade-offs when there are several decision variables and constraints. A systematic algorithm would be much more useful. Better still, a software package. This is discussed in the section after next.

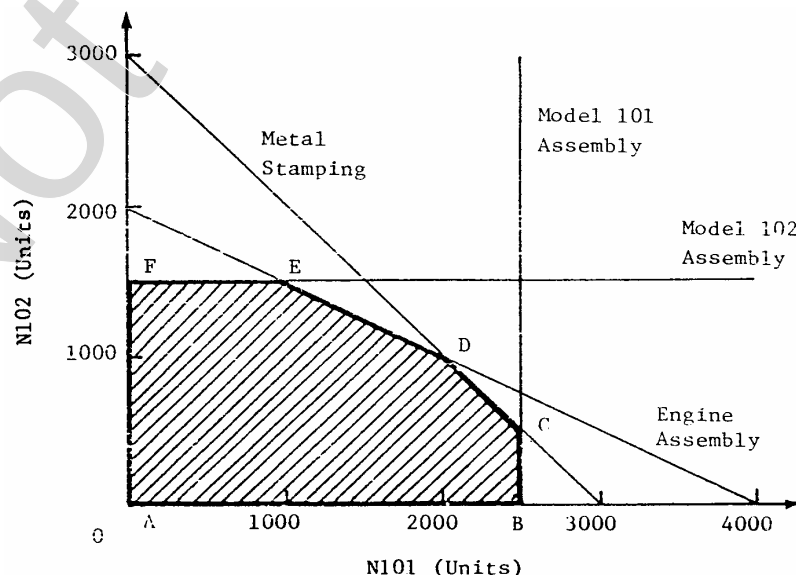
But before that, a graphical approach. **Figure 1** depicts Merton's alternatives on a two-dimensional graph. The "x-axis" represents the number of Model 101 trucks manufactured, while the "y-axis" represents the number of Model 102 trucks. The shaded area is the "feasible set" of production alternatives; it is bounded by straight lines representing the appropriate resource and nonnegativity constraints.

Where in the feasible sets should Merton operate? To obtain the optimal production mix, let us begin with an initial feasible solutions.

The point "A" (0,0) (zero units each of Model 101 trucks and Model 102 trucks) is feasible, but it does not represent a very wise choice: the company's contribution for this production mix is zero, and it can do better by increasing Model 101 production or Model 102 production, or both.

The argument now is almost identical to the previous analysis. Suppose Merton decides to produce Model 101 trucks. Certainly, the company should make the first such truck: it realizes a contribution of \$3000 and it has all this excess production capacity. It should also make a second Model 101 truck. And a third. And a fourth . . . How far can it go? Until it reaches corner point "B" (2500,0). At this stage the monthly production rate for Model 101 is 2500, no Model 102 trucks are being produced, and the monthly contribution is \$7.5 million.

**Figure 1** Merton's Production Possibilities



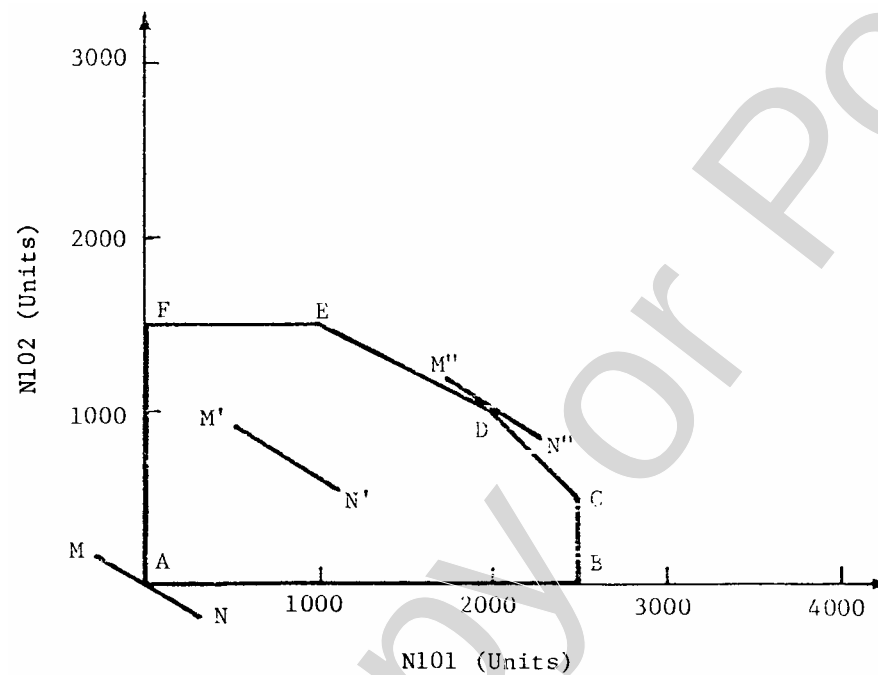
We are now at point "B." Can Merton make further changes in the production mix that increase contribution, but without violating any constraints? The constraint without any slack at "B" is the Model 101 assembly constraint (other than the nonnegativity constraint). Merton can change the production mix by (1) moving back along the line BA (i.e., cutting back Model 101 production); (2) moving in the northwesterly direction inside the feasible set (increasing Model 102 production while cutting back on Model 101 production); or (3) moving northwards along the Model 101 assembly constraint line BC (increasing Model 102 production without cutting back on Model 101 production). Obviously, the third strategy is the best in terms of increased contribution: each Model 102 truck generates an additional \$5000 in contribution. How far should Merton go? It should go on increasing Model 102 production until the corner point "C" is reached. At this stage, the company has run out of metal stamping and Model 101 assembly capacity. The monthly production rates are 2500 Model 101 trucks and 500 Model 102 trucks; and the contribution is \$10 million.

At point "C," Merton can choose to (1) go back along CB (giving up metal stamping capacity), (2) move into the feasible set (giving up both metal stamping and Model 101 assembly capacity), or (3) move along the line CD (giving up Model 101 assembly capacity). Once again, the third strategy is the best one to adopt: by increasing Model 102 production by one unit while at the same time cutting back on Model 101 production by one unit (this trade-off is represented by the metal stamping constraint line CD), Merton can increase contribution by  $\$5000 - \$3000 = \$2000$ . How far can this go on?

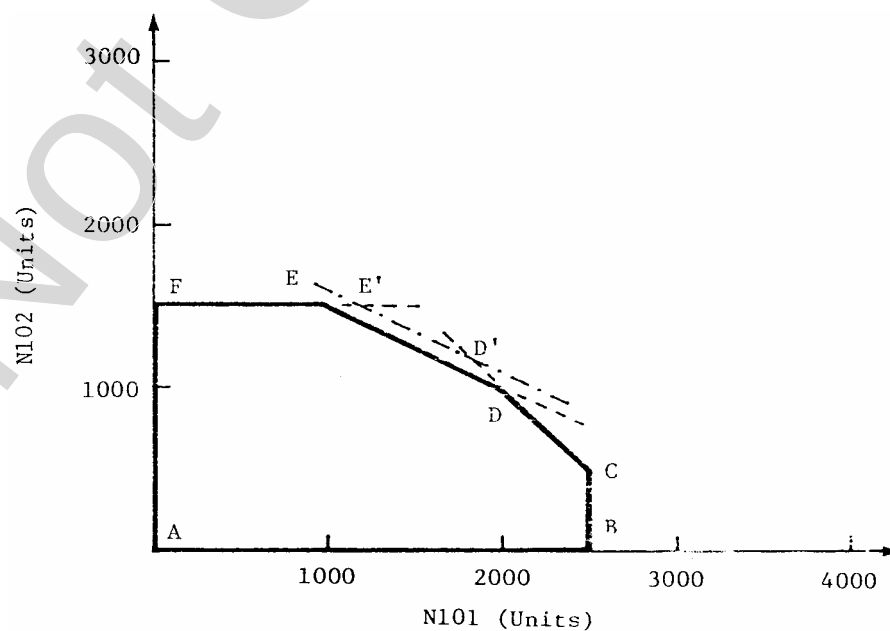
There is no point stopping until the corner point "D" is reached. At this stage, the monthly production rates are 2000 Model 101 trucks and 1000 Model 102 trucks, engine assembly and metal stamping capacities are fully utilized, and the monthly contribution is \$11 million. "D" is indeed the "optimal" point: it is easy to show that a move towards "E," or the interior of the feasible set, or "C" will result in reduced contribution.

An alternative way to get to "D" is indicated in **Figure 2**. The line MN is an iso-contribution line: production mixes at all points on this line will yield the same contribution. MN's slope is  $-3/5$ , the negative of the ratio of the coefficients in the objective function for Model 101 and Model 102. The contribution increases as the line NM moves in a northeasterly direction: the contribution along M'N' is greater than that along MN. It follows that Merton's contribution will be the greatest when NM just touches the feasible set—see M"N", touching the feasible set at D. (In "degenerate" cases M"N" might be parallel to one of the faces of the feasible set, in which case all production mixes along the line will be optimal.)

Having characterized the optimum, we can perform some sensitivity analyses. For example, what would be the consequences of an extra machine-hour of engine assembly capacity? The line DE will move outward and the coordinates of points "D" and "E" will change. The new coordinates of "D" will be (1999,1001) (see **Figure 3**). The optimal strategy is to manufacture 1999 Model 101 trucks and 1001 Model 102 trucks. The increase in contribution is \$2000—the shadow price for the engine assembly constraint. This shadow price will hold until the point "D" overlaps with point "E"—i.e., Model 102 assembly capacity is exhausted. This will happen when the additional engine assembly capacity exceeds 500 machine-hours. Further increase in engine assembly capacity will be wasted because of scarce Model 102 assembly and metal stamping capacity.

**Figure 2** Finding the Optimal Production Mix

Once again, a number of sensitivity analyses are possible. But doing them this way can be cumbersome. In any case, what happens when there are more than two decision variables (and you need more than two dimensions to characterize the feasible set)? An algorithmic or software-based approach would be much better suited to large and repeated problems.

**Figure 3** Sensitivity Analysis: Additional Engine Assembly Capacity

## Sourcing Out Engine Assembly?

We have seen that the shadow price of the engine assembly constraint is \$2000. This means that if engine assembly capacity were to increase by one machine-hour—from 4000 to 4001 machine-hours—Merton could increase its contribution by \$2000. We have also seen that this shadow price holds for up to 500 extra machine-hours.

We conclude that Merton would be willing to pay up to \$2000 per machine-hour of additional engine assembly capacity (over and above the 4000 hours the company already has). Merton should be willing to "buy" up to 500 machine-hours of extra engine assembly capacity at this rate. But no more; any additional capacity is worth zero.

## Using Linear Programming Software

**Exhibit 2** reproduces the output obtained by solving Merton's decision problem with the help of the "CMP" (Conversational Mathematical Programming, Harvard Business School, 1988) Lotus 1-2-3 add-in. (Other linear programming software will give similar outputs.)

The output gives the linear programming "tableau," the optimal values of the decision variables, the optimal value of the objective function, the slacks/surpluses and shadow prices for the different constraints, the ranges of "right-hand side" (resource availability) values for which the shadow prices hold, the "reduced costs" (see next section, "Model 103?"), and the ranges of objective function coefficients for which the reduced costs hold.

It is worth pointing out "complementary slackness" to the students: Leaving aside degenerate cases, when there is a slack (surplus in the case of "greater than or equal to" constraint), the shadow price is zero; and when there is no slack, the shadow price is zero. Similarly, if the optimal solution calls for nonzero levels of the decision variables, the corresponding reduced costs are zero; and when the optimal value for a decision variable is zero, the corresponding reduced costs is nonzero (see next section, "Model 103?").

The linear programming software can be used to investigate the sensitivity of the optimal solution to changes in the problem specification. Consider the case of increased engine assembly capacity. We have argued that the shadow price is \$2000/additional machine-hour for up to 500 machine-hours. This is confirmed in **Exhibits 3, 4, and 5**.

## Model 103?

Merton is considering producing a third model, Model 103. The unit contribution for the new model is \$2000 and each Model 103 truck requires 0.8 (= 4000/5000) machine-hours of engine assembly capacity, 1.5 (= 6000/4000) machine-hours of metal stamping capacity, and 1 machine-hour of Model 101 assembly capacity. Should the company manufacture the new model?

The answer is "No." We can obtain the answer by at least two ways. First, let us try the intuitive approach. Without Model 103, the optimal production mix uses up all the capacity in the engine assembly and metal stamping departments, and there is slack in the Model 101 and Model 102 assembly departments. The shadow prices for the four constraints are \$2000, \$500, \$0, and \$0. If Merton decides to produce one Model 103 truck, it will "divert" 0.8 machine-hours of engine assembly capacity, 1.5 machine-hours of metal stamping capacity, and 1 machine-hour of Model 101 assembly capacity. The cost to the company in lost contribution is  $[(0.8 * 2000) + (1.5 * 500) + (1 * 0) + (0 * 0)] = \$2350$  (the sum of resource use for each constraint times the respective shadow price). The additional contribution from the production and sale of one Model 103 truck is \$2000. Hence, the net impact on contribution is  $-\$2350 + \$2000 = -\$350$ . In other words, Merton's contribution will decrease by \$350 for the first Model 103 truck it manufactures. If the first Model 103 results in a contribution

decrease, many Model 103 trucks certainly will. Conclusion: the company should not manufacture Model 103 trucks.

What we have just done is "priced out" Model 103 trucks and shown that their "reduced cost" is -\$350—requiring a production of one Model 103 truck will reduce contribution by \$350.

To confirm our calculations, let us try a second approach. **Exhibit 6** shows the optimal solution with a column for the decision variable Model 103. The optimal decision is to make zero units of Model 103 and the reduced cost for Model 103 is -\$350. If the contribution for Model 103 were to exceed \$2350 ("High" for Model 103), the optimal solution will change (see **Exhibit 7**). (To convince yourself that this analysis is correct, modify the tableau in **Exhibit 6** with an additional constraint  $N_{103} = 1$ , and observe the contribution decrease by \$350.)

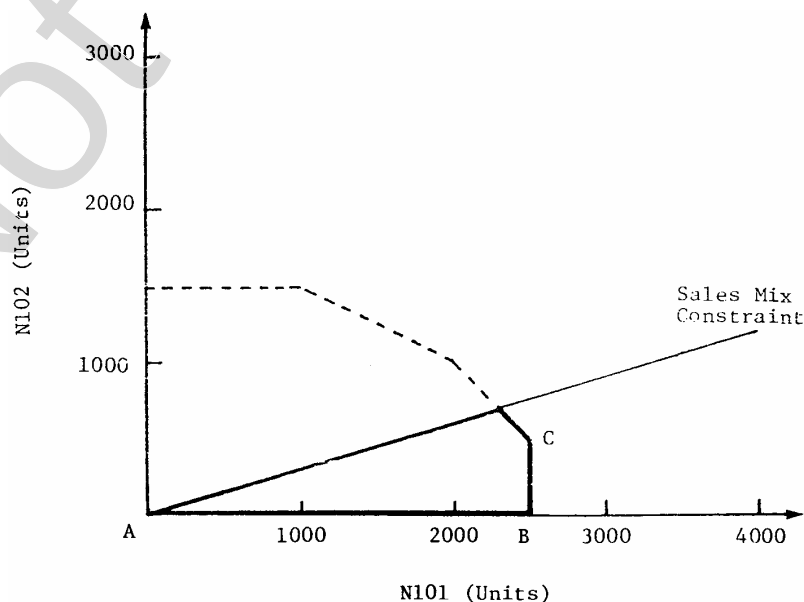
### Overtime Production?

**Exhibit 8** examines the consequences of overtime production in the engine assembly department. There are two more decision variables (the number of Model 101 and Model 102 trucks produced overtime) and one more constraint (engine assembly operations overtime). Note the lower unit contributions in overtime production. We observe that Merton's contribution will increase by \$0.7 million—less than the expected \$0.75 million increase in fixed costs. The conclusion: Merton should not assemble engines overtime.

### Marketing Constraint?

**Exhibit 9** illustrates the consequences of a managerially-imposed sales mix constraint: the number of Model 101 trucks produced must be at least three times the number of Model 102 trucks. The new constraint is indicated graphically in **Figure 4**. The optimal solution is no longer at corner D, but at corner G. The new optimal product mix is 2250 Model 101 trucks and 750 Model 102 trucks; the contribution, at \$10.5 million, is \$0.50 million lower than the original contribution. The result confirms our intuition: a new constraint can only "hurt" (by posing additional restrictions); it cannot "help."

**Figure 4** Adding a Sales Mix Constraint



## Exhibit 1 Board Plan

**Middle Board****#3: Solution**

- One-model case
- Two-model case

**Front Board****#3: Solution**

- Two-model case--graphical approach

**Rear Board****#3: Solution--Sensitivity Analysis**

- Sourcing out engine assembly?
- Model 103?
- Overtime production?
- Marketing constraint?

**Small Left Board****#1: Problem Formulation  
--Conceptual****Small Right Board****#2: Problem Formulation  
--Analytical**

**Exhibit 2** Solution to Merton's Decision Problem CMP Output

|                   | Model101 | Model102 |                   | RHS                    | Slack/Surplus | Shadow Prices | Low  | High    |
|-------------------|----------|----------|-------------------|------------------------|---------------|---------------|------|---------|
| Engine Assembly   | 1        | 2        | <=                | 4000 Engine Assembly   | 0             | 2000          | 3500 | 4500    |
| Metal Stamping    | 2        | 2        | <=                | 6000 Metal Stamping    | 0             | 500           | 5000 | 6500    |
| Model101 Assembly | 2        | 0        | <=                | 5000 Model101 Assembly | 1000          | 0             | 4000 | UNBOUND |
| Model102 Assembly | 0        | 3        | <=                | 4500 Model102 Assembly | 1500          | 0             | 3000 | UNBOUND |
| OBJECTIVE         | 3000     | 5000     | Optimal: 11000000 |                        |               |               |      |         |
|                   | Model101 | Model102 |                   |                        |               |               |      |         |
| Decision Vars     | 2000     | 1000     |                   |                        |               |               |      |         |
| Reduced Costs     | 0        | 0        |                   |                        |               |               |      |         |
| Low               | 2500     | 3000     |                   |                        |               |               |      |         |
| High              | 5000     | 6000     |                   |                        |               |               |      |         |

**Exhibit 3** Increase in Engine Assembly Capacity from 4,000 to 4,001 Machine-hours

|                   | Model101 | Model102 |                   | RHS                    | Slack/Surplus | Shadow Prices | Low  | High    |
|-------------------|----------|----------|-------------------|------------------------|---------------|---------------|------|---------|
| Engine Assembly   | 1        | 2        | <=                | 4001 Engine Assembly   | 0             | 2000          | 3500 | 4500    |
| Metal Stamping    | 2        | 2        | <=                | 6000 Metal Stamping    | 0             | 500           | 5002 | 6501    |
| Model101 Assembly | 2        | 0        | <=                | 5000 Model101 Assembly | 1002          | 0             | 3998 | UNBOUND |
| Model102 Assembly | 0        | 3        | <=                | 4500 Model102 Assembly | 1497          | 0             | 3003 | UNBOUND |
| OBJECTIVE         | 3000     | 5000     | Optimal: 11002000 |                        |               |               |      |         |
|                   | Model101 | Model102 |                   |                        |               |               |      |         |
| Decision Vars     | 1999     | 1001     |                   |                        |               |               |      |         |
| Reduced Costs     | 0        | 0        |                   |                        |               |               |      |         |
| Low               | 2500     | 3000     |                   |                        |               |               |      |         |
| High              | 5000     | 6000     |                   |                        |               |               |      |         |

**Exhibit 4** Increase in Engine Assembly Capacity from 4,000 to 4,100 Machine-hours

|                   | Model101 | Model102 |                   | RHS                    | Slack/Surplus | Shadow Prices | Low  | High    |
|-------------------|----------|----------|-------------------|------------------------|---------------|---------------|------|---------|
| Engine Assembly   | 1        | 2        | <=                | 4100 Engine Assembly   | 0             | 2000          | 3500 | 4500    |
| Metal Stamping    | 2        | 2        | <=                | 6000 Metal Stamping    | 0             | 500           | 5200 | 6600    |
| Model101 Assembly | 2        | 0        | <=                | 5000 Model101 Assembly | 1200          | 0             | 3800 | UNBOUND |
| Model102 Assembly | 0        | 3        | <=                | 4500 Model102 Assembly | 1200          | 0             | 3300 | UNBOUND |
| OBJECTIVE         | 3000     | 5000     | Optimal: 11200000 |                        |               |               |      |         |
|                   | Model101 | Model102 |                   |                        |               |               |      |         |
| Decision Vars     | 1900     | 1100     |                   |                        |               |               |      |         |
| Reduced Costs     | 0        | 0        |                   |                        |               |               |      |         |
| Low               | 2500     | 3000     |                   |                        |               |               |      |         |
| High              | 5000     | 6000     |                   |                        |               |               |      |         |

**Exhibit 5** Increase in Engine Assembly Capacity from 4,000 to 4,501 Machine-hours

|                   | Model101 | Model102 |                   | RHS                    | Slack/Surplus | Shadow Prices | Low  | High    |
|-------------------|----------|----------|-------------------|------------------------|---------------|---------------|------|---------|
| Engine Assembly   | 1        | 2        | <=                | 4501 Engine Assembly   | 1             | 0             | 4500 | UNBOUND |
| Metal Stamping    | 2        | 2        | <=                | 6000 Metal Stamping    | 0             | 1500          | 3000 | 6002    |
| Model101 Assembly | 2        | 0        | <=                | 5000 Model101 Assembly | 2000          | 0             | 3000 | UNBOUND |
| Model102 Assembly | 0        | 3        | <=                | 4500 Model102 Assembly | 0             | 666.6666      | 1500 | 4503    |
| OBJECTIVE         | 3000     | 5000     | Optimal: 12000000 |                        |               |               |      |         |
|                   | Model101 | Model102 |                   |                        |               |               |      |         |
| Decision Vars     | 1500     | 1500     |                   |                        |               |               |      |         |
| Reduced Costs     | 0        | 0        |                   |                        |               |               |      |         |
| Low               | 0        | 3000     |                   |                        |               |               |      |         |
| High              | 5000     | UNBOUND  |                   |                        |               |               |      |         |

## Exhibit 6 Model 103?

|                   | Model101 | Model102 | Model103 |                   | RHS                    | Slack/Surplus | Shadow Prices | Low  | High    |
|-------------------|----------|----------|----------|-------------------|------------------------|---------------|---------------|------|---------|
| Engine Assembly   | 1        | 2        | 0.8      | <=                | 4000 Engine Assembly   | 0             | 2000          | 3500 | 4500    |
| Metal Stamping    | 2        | 2        | 1.5      | <=                | 6000 Metal Stamping    | 0             | 500           | 5000 | 6500    |
| Model101 Assembly | 2        | 0        | 1        | <=                | 5000 Model101 Assembly | 1000          | 0             | 4000 | UNBOUND |
| Model102 Assembly | 0        | 3        | 0        | <=                | 4500 Model102 Assembly | 1500          | 0             | 3000 | UNBOUND |
| OBJECTIVE         | 3000     | 5000     | 2000     | Optimal: 11000000 |                        |               |               |      |         |
|                   | Model101 | Model102 | Model103 |                   |                        |               |               |      |         |
| Decision Vars     | 2000     | 1000     | 0        |                   |                        |               |               |      |         |
| Reduced Costs     | 0        | 0        | -350     |                   |                        |               |               |      |         |
| Low               | 2500     | 3000     | UNBOUND  |                   |                        |               |               |      |         |
| High              | 5000     | 6000     | 2350     |                   |                        |               |               |      |         |

## Exhibit 7 Model 103's Contribution Increased to \$2,351

|                   | Model101 | Model102 | Model103 |                   | RHS                    | Slack/Surplus | Shadow Prices | Low      | High    |
|-------------------|----------|----------|----------|-------------------|------------------------|---------------|---------------|----------|---------|
| Engine Assembly   | 1        | 2        | 0.8      | <=                | 4000 Engine Assembly   | 0             | 1998.571      | 3200     | 4600    |
| Metal Stamping    | 2        | 2        | 1.5      | <=                | 6000 Metal Stamping    | 0             | 501.4285      | 4857     | 7500    |
| Model101 Assembly | 2        | 0        | 1        | <=                | 5000 Model101 Assembly | 2142.857      | 0             | 2857.142 | UNBOUND |
| Model102 Assembly | 0        | 3        | 0        | <=                | 4500 Model102 Assembly | 1928.571      | 0             | 2571.428 | UNBOUND |
| OBJECTIVE         | 3000     | 5000     | 2351     | Optimal: 11002857 |                        |               |               |          |         |
|                   | Model101 | Model102 | Model103 |                   |                        |               |               |          |         |
| Decision Vars     | 0        | 857.1428 | 2857.142 |                   |                        |               |               |          |         |
| Reduced Costs     | -1.42857 | 0        | 0        |                   |                        |               |               |          |         |
| Low               | UNBOUND  | 3134.666 | 2350     |                   |                        |               |               |          |         |
| High              | 3001.428 | 5020     | 3750     |                   |                        |               |               |          |         |

## Exhibit 8 Overtime Production?

|                   | Mod101Reg | Mod102Reg | Mod101OT | Mod102OT | RHS                       | Slack/Surplus | Shadow Prices | Low  | High    |
|-------------------|-----------|-----------|----------|----------|---------------------------|---------------|---------------|------|---------|
| Eng Ass-Regular   | 1         | 2         | 0        | 0        | <= 4000 Engine Assembly   | 0             | 600           | 2500 | 4500    |
| Metal Stamping    | 2         | 2         | 2        | 2        | <= 6000 Metal Stamping    | 0             | 1200          | 5000 | 8000    |
| Model101 Assembly | 2         | 0         | 2        | 0        | <= 5000 Model101 Assembly | 2000          | 0             | 3000 | UNBOUND |
| Model102 Assembly | 0         | 3         | 0        | 3        | <= 4500 Model102 Assembly | 0             | 466.6666      | 3000 | 9000    |
| Eng Ass-Overtime  | 0         | 0         | 1        | 2        | 2000 Eng Ass-Overtime     | 1500          | 0             | 500  | UNBOUND |
| OBJECTIVE         | 3000      | 5000      | 2400     | 3800     | Optimal: 11700000         |               |               |      |         |
|                   | Mod101Reg | Mod102Reg | Mod101OT | Mod102OT |                           |               |               |      |         |
| Decision Vars     | 1500      | 1250      | 0        | 250      |                           |               |               |      |         |
| Reduced Costs     | 0         | 0         | -4,5E-13 | 0        |                           |               |               |      |         |
| Low               | 3000      | 3800      | UNBOUND  | 3800     |                           |               |               |      |         |
| High              | 4400      | 5000      | 2400     | 5000     |                           |               |               |      |         |

## Exhibit 9 Marketing Constraint?

|                   | Model101 | Model102 | RHS                       | Slack/Surplus | Shadow Prices | Low   | High    |
|-------------------|----------|----------|---------------------------|---------------|---------------|-------|---------|
| Engine Assembly   | 1        | 2        | <= 4000 Engine Assembly   | 250           | 0             | 3750  | UNBOUND |
| Metal Stamping    | 2        | 2        | <= 6000 Metal Stamping    | 0             | 1750          | 0     | 6400    |
| Model101 Assembly | 2        | 0        | <= 5000 Model101 Assembly | 500           | 0             | 4500  | UNBOUND |
| Model102 Assembly | 0        | 3        | <= 4500 Model102 Assembly | 2250          | 0             | 2250  | UNBOUND |
| Sales Mix         | 1        | -3       | >= 0 Sales Mix            | 0             | -500          | -1000 | 1000    |
| OBJECTIVE         | 3000     | 5000     | Optimal: 10500000         |               |               |       |         |
|                   | Model101 | Model102 |                           |               |               |       |         |
| Decision Vars     | 2250     | 750      |                           |               |               |       |         |
| Reduced Costs     | 0        | 0        |                           |               |               |       |         |
| Low               | -1666.66 | 3000     |                           |               |               |       |         |
| High              | 5000     | UNBOUND  |                           |               |               |       |         |